PassFrame
tuition

Ratio & Proportion

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Share £36 in the ratio 1:3.
Method & answer: 4 parts. 1 part = £9. Shares: £9 and £27.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Simplify the ratio 6:12.
2. Simplify the ratio 14:21.
3. Simplify the ratio 10:12.
4. Share 15 sweets in the ratio 1:2.
5. Share 25 marbles in the ratio 1:4.
6. Share 27 in the ratio 1:2.
7. Simplify the ratio 18:24.
8. Write the ratio of 4 pens to 12 pencils in simplest form.
9. Share 35 in the ratio 2:3.
10. Simplify the ratio 12:30.
11. Share 32 in the ratio 1:3.
12. Write the ratio 20:35 in simplest form.
13. Share 48 in the ratio 3:5.
14. Simplify the ratio 15:20.
15. Share 28 in the ratio 2:5.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Share £120 in the ratio 3:5.
2. Share 144kg in the ratio 2:3:7.
3. A ratio is 4:7. If the smaller amount is 32, what is the larger?
4. A ratio is 3:8. If the larger amount is 120, what is the smaller?
5. Share 180 in the ratio 3:4:5.
6. If 6 pens cost £5.40, how much do 15 pens cost?
7. If 5 books cost £27.50, how much do 8 books cost?
8. A map scale is 1:250. A distance of 8cm on the map is how many metres in real life?
9. For every 3 boys in a class there are 5 girls. If there are 24 boys, how many girls?
10. Simplify the ratio 36:108.
11. Share 130 in the ratio 4:9.
12. A ratio is 7:9. If the total is 160, find both parts.
13. Write 60p to £3 as a ratio in its simplest form.
14. If 8kg of oranges costs £18, how much do 12kg cost?
15. Share 252 in the ratio 4:3:7.

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. 6 taps fill a tank in 4 hours. How long for 8 taps?
6. A fruit punch is made from orange juice, lemonade and soda in the ratio 3:2:1. If 4.8 litres of punch are made in total, how many litres
11. A sum of £198 is shared between three friends in the ratio 2:3:4. How much does the friend with the largest share receive?

of lemonade are used?

2. A ratio is 2:3:4. If the difference between the largest and smallest share is 24, find the total.

3. Simplify 0.4 : 1.6.

4. A ratio 3:5 is increased so that the second number becomes 45. What is the new first number, keeping the ratio the same?

5. Share £60 in the ratio 1:2:3, then find how much more the largest share gets than the smallest.

7. Two friends, Jack and Mia, invest money in a business in the ratio 5:3. If Jack invests £2000 more than Mia, how much did each invest?

8. A recipe for 6 people uses 450g of pasta. How much pasta is needed to serve 15 people?

9. A school has pupils and teachers in the ratio 15:1. If there are 480 pupils, how many teachers are there?

10. A car uses 8 litres of petrol to travel 120km. How far can it travel on a full tank of 50 litres, at the same rate?

12. A garden has red and yellow flowers in the ratio 5:7. There are 20 more yellow flowers than red. How many flowers are there in total?

13. A photo printed at actual size 10cm by 15cm is scaled down by a ratio of 2:5. What are the new dimensions?

14. 3 identical taps fill a paddling pool in 40 minutes. If 5 taps are used instead, how long will it take?

15. A necklace uses blue, red and gold beads in the ratio 4:3:2. If there are 63 beads in total that are NOT gold, how many gold beads are there?